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Key Points:

- SSA-PD for solving the phase shift phenomenon is proposed
- SSA-P for predicting coordinates of a GPS site is developed
- Enhanced SSA can analyze and predict geodetic observations affected by several phenomena

Correspondence to:

Y. Cheng and X. Wang,
 yycheng@casm.ac.cn;
 wxmgeo@gmail.com

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An enhanced singular spectrum analysis method for constructing nonsecular model of GPS site movement

Xiaoming Wang¹, Yingyan Cheng², Suqin Wu¹, and Kefei Zhang¹
¹SPACE Research Centre, School of Mathematical and Geospatial Sciences, RMIT University, Melbourne, Victoria, Australia,

²Chinese Academy of Surveying and Mapping, Beijing, China

Abstract Many GPS time series contain offsets, sometimes nonsecular trends, and seasonal signals with time-varying amplitudes due to several different types of geophysical phenomena. Therefore, the use of nonsecular models to depict the real geophysical movement of GPS sites is better than a linear model. In this study, an enhanced singular spectrum analysis (SSA) method for fitting GPS time series and predicting its coordinates is proposed. Simulation results show that the root-mean-square (RMS) of differences between the reconstructed and simulated signals is 1.7 mm; the RMS of the differences between the predicted coordinates and simulated signal is about 3 mm for the first half 1.5 years of testing period and decreases to 10 mm for the last half 1.5 years. Fitting results for three GPS time series are obtained using maximum likelihood estimation (MLE), which is used to fit the time series with a piecewise linear trend plus an annual/semiannual components, SSA, and state space model (SSM). Both SSA and SSM perform similarly and better than the MLE in extracting the nonsecular trend and annual/semiannual components from the GPS time series. The prediction results from SSA have higher coefficients with raw time series and lower power of annual/semiannual in their residuals than that from MLE for two case studies. The differences between the linear trend estimated by Plate Boundary Observatory and SSA nonsecular model for 16 GPS time series are all larger than 2 mm in up direction, which is not negligible for a high-accuracy terrestrial reference frame construction.

1. Introduction

A terrestrial reference system (TRS) is defined by three fundamental elements: origin, scale, and orientation, with other specific physical and mathematical properties of the system [Altamimi *et al.*, 2007, 2011]. It is a conventional reference system and realized through a terrestrial reference frame (TRF), which is constructed with space observations from globally distributed geodetic reference stations. Ground-based GPS stations, together with other geodetic observations, e.g., very long baseline interferometry, have been used to establish a TRF [Altamimi and Collilieux, 2009]. For the establishment of a TRF, a linear model has always been used to fit the velocity of the geodetic reference sites [Altamimi *et al.*, 2012].

Several studies [Williams, 2003b; Gazeaux *et al.*, 2013; Montillet *et al.*, 2015] pointed out that site velocities estimated from their time series can be biased, if offsets exist and they are not corrected. Large offsets can be detected through a mathematical method or by manual methods. However, small offsets can hardly be detected by these two methods. To address this issue, a community experiment for detecting offsets in GPS time series, The Detection of Offsets in GPS Experiment (DOGEx), was initiated by Gazeaux *et al.* in 2010 [Gazeaux *et al.*, 2013]. The first results from this experiment showed that manual methods (where offsets were handpicked) almost always gave better results than automated or semi-automated methods. Moreover, previous studies also indicated that the movement of a site is often nonsecular in the cases of postseismic deformation or deformation associated with rapid elastic or poroelastic phenomenon. For example, Tregoning *et al.* [2013] identified a small consistent nonsecular postseismic relaxation deformation in GPS time series along the eastern coast of Australia. This research implied that the presence of ongoing postseismic relaxation violates the linear assumption made in the current TRF definition. This linear assumption has significant ramifications for regional tectonic interpretations and global studies such as those focusing on sea level rise which require a level of 1 mm accuracy of TRF [Gross *et al.*, 2009]. In addition, previous studies [Bennett, 2008; Davis *et al.*, 2012; Chen *et al.*, 2013] also indicated that the amplitudes and phases of seasonal signals detected from many GPS time series are time varying.

It is important and also challenging to construct a nonsecular model to depict the real geophysical movement of GPS sites from their time series if the data contain offsets, nonsecular trends, and periodic components with time-varying amplitudes. Previous research focused on improving the accuracy of linear velocity estimates rather than using nonsecular models to characterize the real movement of GPS sites, even the latest international TRF (ITRF) was realized based on a linear model [Altamimi *et al.*, 2007; Altamimi and Collilieux, 2009; Altamimi *et al.*, 2012; Argus, 2012]. Hence, Tregoning *et al.* [2013] pointed out that the current ITRF could not depict the real movement of GPS sites.

This research aims to develop an improved method for the construction of a nonsecular TRF model. Three methods for time series analysis are investigated. The first method is singular spectrum analysis (SSA). It is a data-driven technique that uses time domain data to extract information from noisy time series without use of prior knowledge of physical phenomena in the time series [Vautard and Ghil, 1989; Elsner and Tsonis, 1996; Hassani, 2007; Chen *et al.*, 2013]. In recent years, SSA has also been used in geodetic study and one of the latest applications is the modeling of seasonal signals from GPS time series [Chen *et al.*, 2013]. However, one limitation of SSA is the large biases in the beginning and end of its model fitting to a time series, which is called phase phenomenon [Vautard and Ghil, 1989; Vautard *et al.*, 1992].

To address the phase shift issue of SSA in our study a new method named SSA with pseudo data (SSA-PD) is investigated. Another expanded method called SSA for prediction (SSA-P) is proposed for predicting the coordinates of GPS sites at any specific time. Their performances are assessed using both simulated and real GPS time series. To illustrate the advantages of using the enhanced SSA-PD in time series analysis, comparisons among three GPS time series fitting results obtained by SSA-PD, maximum likelihood estimation (MLE) [Langbein and Johnson, 1997; Zhang *et al.*, 1997], and state space model (SSM) [Laine *et al.*, 2014] are conducted. When the MLE method is used, the information of the offset epochs is used. This offset information is not used for the SSA and SSM methods. In this study, the dynamic linear model (DLM) MATLAB toolbox (based on the SSM method) developed by Laine [Laine *et al.*, 2014] and the Create and Analyse Time Series (CATS) software (based on the MLE method) developed by Williams, [Williams, 2008] are adopted. Since the DLM toolbox does not provide a prediction function, the prediction results only from MLE and SSA are compared. For the MLE method, a piecewise linear trend plus an annual/semiannual component is applied to fit the GPS time series.

2. Methodology and Simulation Result

In this section the main theory of the MLE, SSM, and SSA methods is introduced and the two proposed methods (SSA-PD and SSA-P) are explained with the help of simulated time series.

2.1. MLE

The movement of a GPS site is commonly modeled by

$$y_{t_i} = y_0 + v(t_i - t_0) + \sum_{p=1}^2 a_p (\sin 2\pi f_p t_i + \omega_p) + \sum_{j=1}^n \Delta y_j H(t_i - t_j) + \varepsilon_{t_i} \quad (1)$$

where y_{t_i} is the position of the GPS site at epoch t_i , y_0 is the position at the initial time t_0 , v is the linear velocity, a_p and ω_p are the amplitude and phase of each seasonal signal at frequency f_p ($p=1$ for annual component and $p=2$ for semiannual component) respectively, $\sum_{j=1}^n \Delta y_j H(t_i - t_j)$ is the Heaviside function defining the coordinate jump (Δy_j) at time t_j , and ε_{t_i} is the noise.

It should be noted that a postseismic exponential or logarithmic logarithm is sometimes included to account for the site movement after earthquakes occur.

Several studies concluded that an optimal model for characterizing the noise in GPS time series is a combination of white and power law noises [Mao *et al.*, 1999; Williams, 2003a; Amiri-Simkooei *et al.*, 2007; Santamaría-Gómez *et al.*, 2011; Hackl *et al.*, 2011, 2013; He *et al.*, 2015]. The covariance matrix of the time series measurements then can be expressed as

$$C = \alpha^2 I + \beta^2 J^2 \quad (2)$$

where α and β are the amplitudes of the white and flicker noise, respectively, I is the identity matrix, and J is the covariance matrix of the flicker noise.

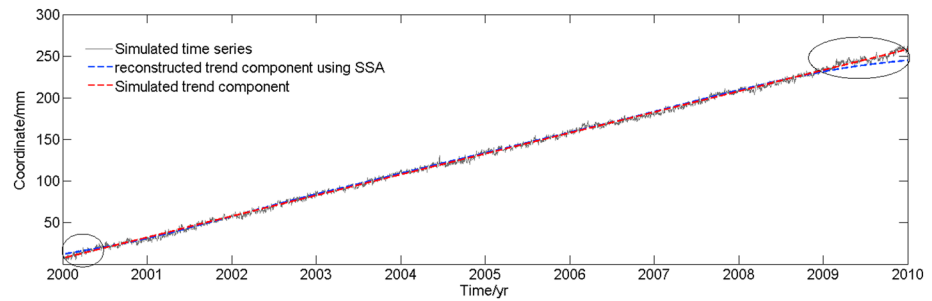


Figure 1. Reconstructed trend component.

The CATS software package adopted in this study uses least squares to fit a multiparameter model to a time series while concurrently analyzing the residuals for assessing the type and magnitude of the stochastic noise [Williams, 2008]. The MLE implemented in CATS is for the assessment of the noise content in a GPS time series. More details about CATS can be found in Williams [2008]. Instead of using CATS, one can also use the Hector software of Bos *et al.* [2013] which is faster than CATS.

2.2. SSM

The SSM model, sometimes called hidden Markov models or structural time series models, are well known and introduced in the literature [Laine *et al.*, 2014].

A general linear SSM model with Gaussian errors can be expressed by an observation equation and a state evolution equation as

$$\begin{aligned} y_{t_i} &= F_t \cdot x_{t_i} + v_{t_i} & v_{t_i} &\sim N_p(0, V_{t_i}) \\ x_{t_i} &= G_{t_i} \cdot x_{t_{i-1}} + w_{t_i} & w_{t_i} &\sim N_q(0, W_{t_i}) \end{aligned} \quad (3)$$

where y is the observation vector of length p , x is a vector containing state variables of length q , F is the observation operator mapping the states to the observations, and G is the state evolution operator giving the dynamics of the states.

In the above formulae, the uncertainties v and w are assumed to be Gaussian, with observation uncertainty covariance V and model error covariance W . The G , F , and W are assumed to be time invariant, while V is time dependent. It is useful to have some prior information for determining W .

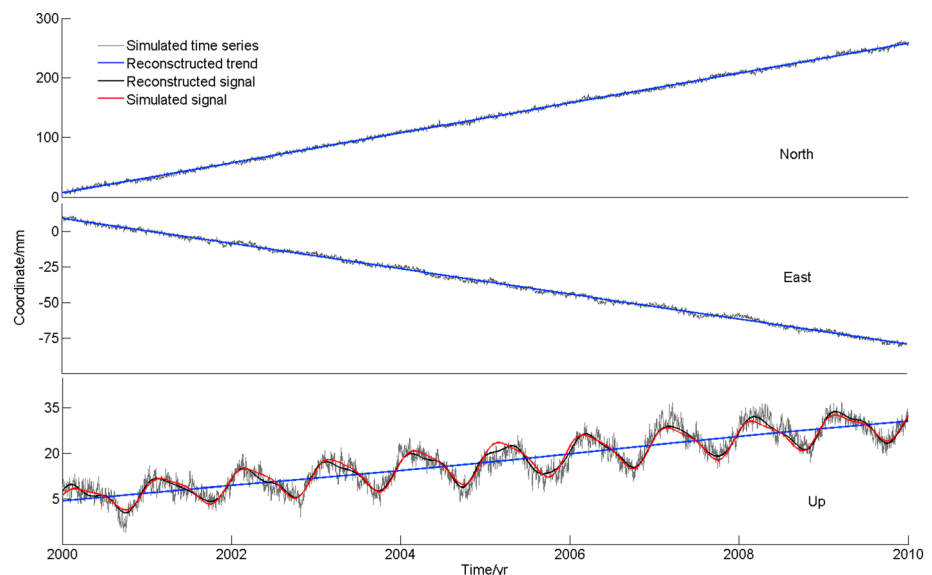


Figure 2. Reconstructed signals and trend components using SSA-PD.

Table 1. Position Epochs and Magnitudes of Offsets

	Epoch (year)	North (mm)	East (mm)	Up (mm)
1	2001.645	3	6	-4
2	2003.835	-23	8	18
3	2005.478	2	-12	-3
4	2007.669	-12	7	-5

2.3. SSA

The procedure of SSA mainly consists of two stages: decomposition and reconstruction [Vautard and Ghil, 1989; Vautard et al., 1992; Elsner and Tsonis, 1996] and more details can be found in Appendix A. A key problem of using SSA is the selection of an optimal lag window M . As suggested by Vautard et al. [1992], the M value should be neither too small nor too large. A small M may lead to several neighboring peaks in the spectrum of a periodic component to coalesce due to the coarse resolution. In contrast, a large M value may split the peaks into several components with neighboring frequencies. In the geodetic field, Rangelova et al. [2012] applied a 3 year lag window to extract the long-term trend together with seasonal variations from GRACE time series. Chen et al. [2013] suggested that 2 year or 3 year lag window seem to be appropriate for extracting the annual and semi-annual components. However, if long-term variations are of interest, a larger M value should be adopted [Rangelova et al., 2012; Chen et al., 2013]. In this study, a 3 year lag window is used for extracting a long-term trend first to obtain a detrended time series and then a 2 year lag window is applied to extract seasonal signals from the detrended time series. In addition, the enhanced method SSA-PD to address the phase shift issue of SSA is detailed in Appendix B1 and another method named SSA-P for predicting the coordinates of GPS sites at any specific time is detailed in Appendix B2. The software package we used for analyzing the time series of a GPS site (based on SSA-PD) and for predicting its coordinates at any epochs (based on SSA-P) is available at <https://sites.google.com/site/ssasoftwarev1/>. We have also provided a brief description at the website about the software package and some GPS data for showing the input data format.

3. Simulation Results

3.1. SSA-PD

To explain the phase shift phenomenon and the proposed SSA-PD method, three types of time series are simulated and the performance of SSA-PD for the modeling of the trend and periodic components from these time series are assessed. The first type of data consists of a linear trend, two periodic components, and a combination of flicker noise and white noise. The second type of time series is simulated by adding four offsets into the first type of data series. The third type of data consists of a nonsecular component, four offsets, two periodic components, and a combination of flicker noise and white noise.

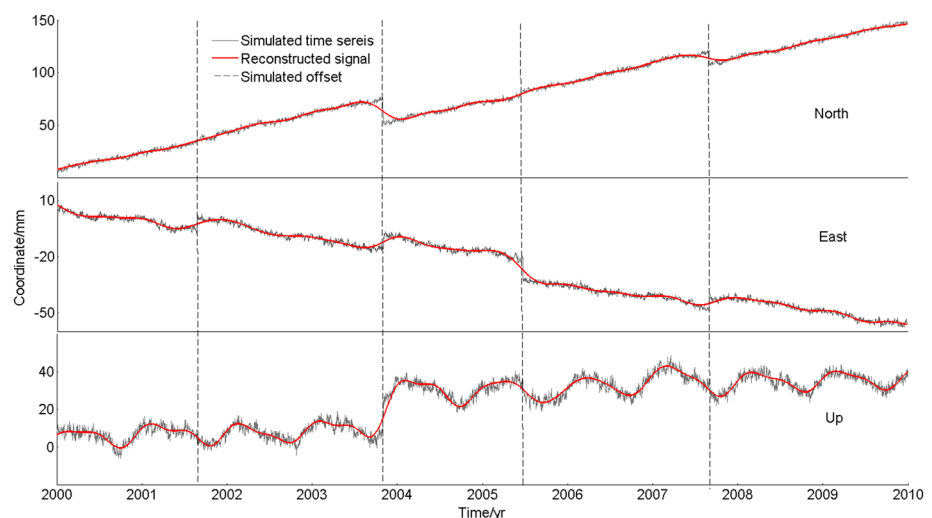


Figure 3. Reconstructed signal component using SSA-PD from time series containing offsets.

Table 2. Fitting Accuracy of the Time Series Before and After an Offset

Offset Epoch	Fitting Accuracy Before Offset (mm)	Fitting Accuracy After Offset (mm)	Offset Epoch	Fitting Accuracy Before Offset (mm)	Fitting Accuracy After Offset (mm)
2009.500	1.5	3.6	2007.000	1.6	2.7
2009.000	1.7	3.3	2006.000	1.6	1.9
2008.500	1.8	1.8	2005.500	1.4	1.9
2008.000	2.2	3.7	2005.000	1.4	1.7
2007.500	2.4	3.0	2004.500	1.9	0.5

The formula for the first type of data simulation is as follows:

$$y_{t_i} = x_{t_i} + v_{t_i} \quad (4)$$

where x is the signal and v is the noise, and x can be simulated by

$$x_{t_i} = x_{t_0} + v(t_i - t_0) + (c_{t_i} + a) \cdot \sin(2\pi \cdot t_i + \varphi_1) + (d_{t_i} + b) \cdot \sin(4\pi \cdot t_i + \varphi_2) \quad (5)$$

where t_i is the epoch of i th observation in decimal year, x_{t_0} is the intercept at the initial epoch t_0 , v is the velocity of the linear trend, a and b are constant amplitudes, c_t and d_t are time-varying amplitudes, and φ_1 and φ_2 are two phase parameters.

Previous studies [Bennett, 2008; Davis et al., 2012; Chen et al., 2013] stated that the seasonal signals in GPS time series were modulated. Therefore, we choose the amplitudes of the annual and semiannual signals to vary according to

$$c_{t_i} = 0.7a \cdot \sin\left(\frac{i}{400 \cdot \pi}\right) \quad (6)$$

$$d_{t_i} = 0.5b \cdot \sin\left(\frac{i}{1000 \cdot \pi}\right) \quad (7)$$

The value of the velocity v in equation (5) is randomly chosen from a reasonable distribution of earth surface velocities, and a and b are randomly chosen from a reasonable range based on the estimate results of the annual and semiannual signals from real GPS data provided by the Scripps Orbit and Permanent Array Center (SOPAC). The $\sin$ functions are used to define the variation of the amplitudes, and the two constants adopted in the functions are 400 and 1000.

The noise v_{t_i} in equation (4) is simulated using a combination of white and flicker noises based on the state-of-the-art studies [Williams, 2003a; Gazeaux et al., 2013]. The magnitudes of white and flicker noises are randomly chosen from a reasonable range based on the real noise magnitude results in these previous studies.

The reconstructed trend component using SSA and the simulated data is shown in Figure 1. The biases between the reconstructed and simulated trends at the beginning and end of the time series are noticeable.

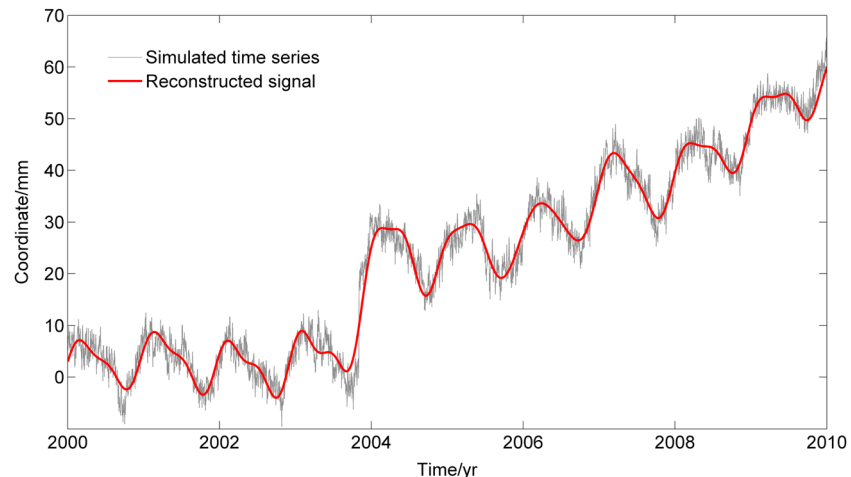


Figure 4. Reconstructed signal component f SSA-PD from time series containing offsets and a nonsecular trend.

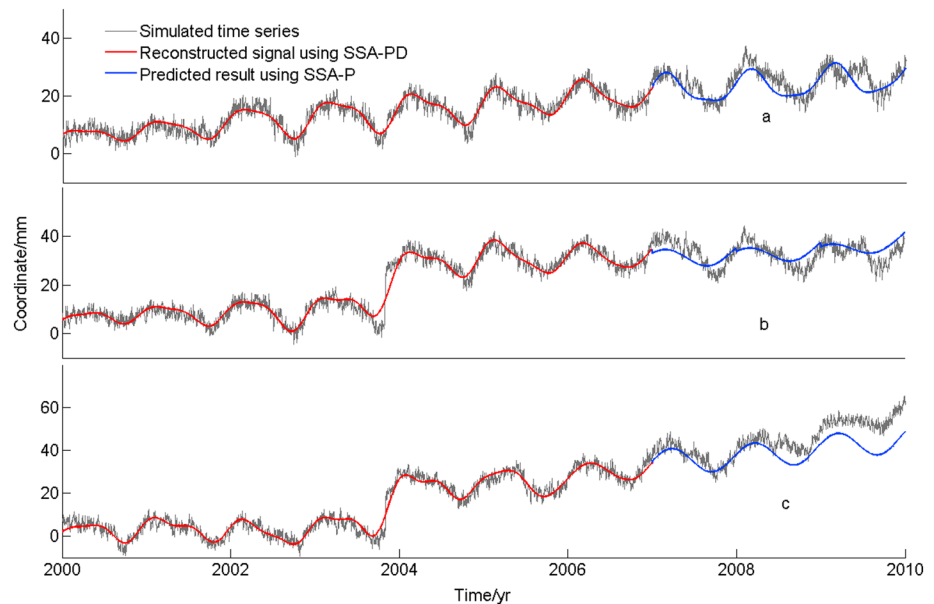


Figure 5. Prediction result from SSA-P for the (a) first type of data, (b) second type of data, and the (c) third type of data.

These biases are called phase shift and they are caused by the fact that different formulae are used to calculate different subsets in the RCs (reconstruct components). *Chen et al.* [2013] also stated that the presence of phase shift in the reconstructed trend component could adversely affect the reconstructed periodic components due to the aliasing of some RCs. The SSA-PD method proposed in this study is for solving this phase shift problem so that a more reliable trend component can be reconstructed.

To test the performance of SSA-PD in trend reconstructing, a 3-D time series is simulated and the reconstructed results are shown in Figure 2, in which no phase shift appears in both beginning and end parts of the reconstructed trend.

To assess the performance of SSA-PD for time series containing unexpected offsets, a new 3-D time series containing offsets is simulated. According to the SOPAC [Gazeaux et al., 2013], the magnitudes of most offsets in GPS time series are under 25 mm. This is why the magnitude of the offsets simulated in this study are randomly selected from the range -25 – 25 mm. Apart from this, if the offsets are larger than 25 mm, they can be easily detected or picked. The epochs and magnitudes of the offsets are shown in Table 1, and the reconstructed signal from SSA-PD is shown in Figure 3. The RMS (root-mean-square) of the differences between the reconstructed and simulated signals (trend plus periodic components) are 1.2 mm, 1.3 mm, and 1.6 mm in the north, east and up directions, respectively. It can be also found from Figure 3 that the fitting results are not very good around the offset epoch. Since no offset information is used when applying

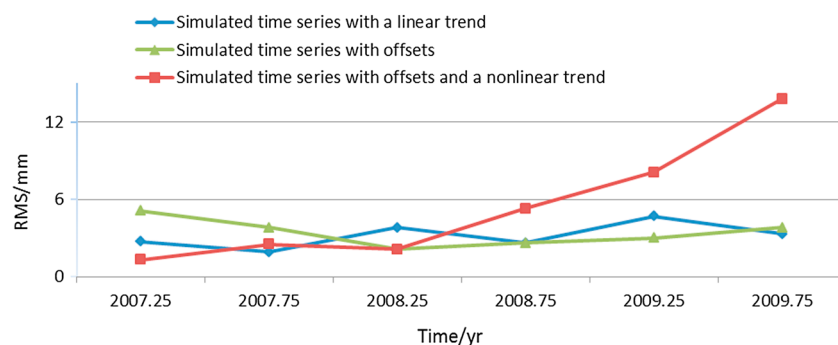


Figure 6. RMS of the differences between the predicted and simulated signals for the period ± 0.25 year either side of the time label.

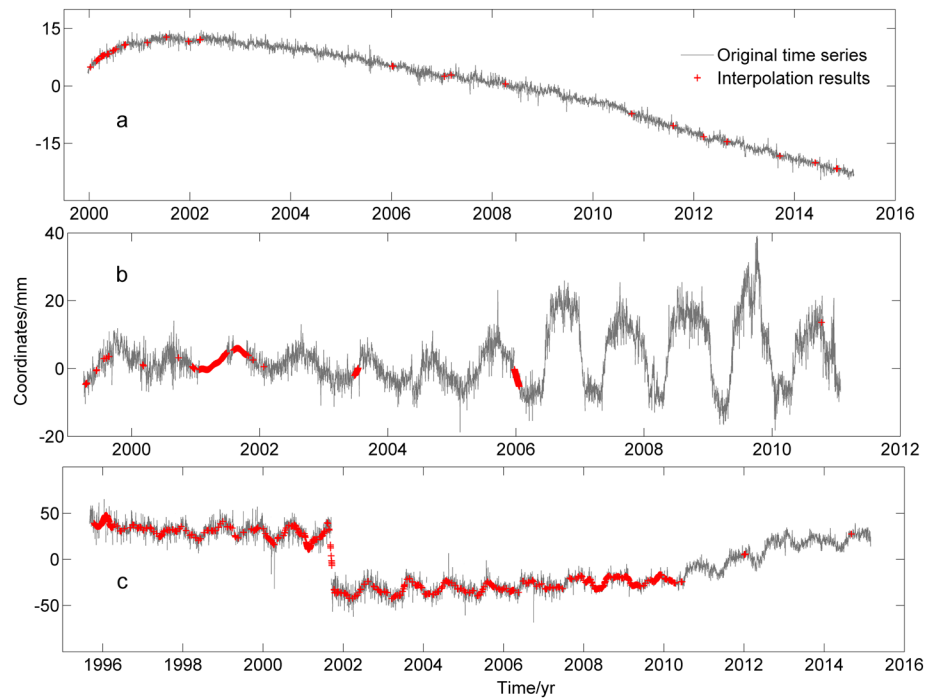


Figure 7. Interpolation results for time series of (a) AGMT in the north direction, (b) DSHS in the vertical direction and (c) KELY in the vertical direction.

SSA-PD to fit the time series, no optimal method for reducing the fitting error around the offset epoch is proposed in this study.

To test the sensitivity of this approach to the location of the offset, new time series are simulated by moving the largest offset in Figure 3 (up) from the epoch 2009.500 to 2005.000 by a step of 0.5 years. The fitting accuracy of the time series before and after the offset is listed in Table 2. The results show that the fitting accuracy of the time series after the offset is sensitive to its location. When the large offset appears within the 4 years before the end of the series, the RMS values between the simulated and constructed signals after the offset are larger than 2 mm. Therefore, if the time series contains a large offset within the last few years (e.g., 4 years) and new GPS data are available (e.g., 1 year), it should be reprocessed using SSA to improve the fitting accuracy.

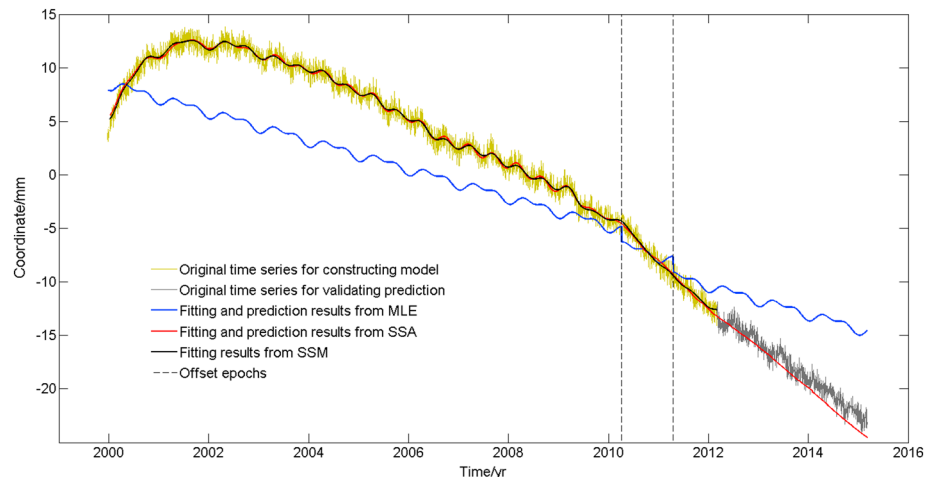


Figure 8. Signal extracted by MLE, SSA-PD, and SSM from the vertical coordinate of AGMT for 1999.975–2015.18.

Table 3. Correlation Coefficients Between the Original Data and Fitting Results From SSA-PD, SSM, and MLE for the Vertical Coordinates of AGMT for 1999.975–2012.167

	Original Data	SSA-PD	SSM	MLE
Original data	1			
SSA-PD	0.99	1		
SSM	0.99	0.99	1	
MLE	0.96	0.96	0.96	1

To assess the performance of SSA-PD for time series containing a nonsecular trend and offsets, a new time series is simulated by replacing the linear trend in the time series (the up direction) in Figure 3 with a nonsecular trend by

$$T_i = e^{(0.3^*(t_i - 2000.0) + 1)} \quad (8)$$

where t_i is the epoch of the i th observation in decimal year.

The reconstructed signal is shown in Figure 4. The RMS of the differences between the simulated and reconstructed signals is 1.7 mm.

3.2. SSA-P

The SSA-P method for predicting the coordinates of GPS sites at any epoch is introduced in Appendix B2, and its performance is assessed using simulated time series in this section.

To assess SSA-P, three types of data with a length of 10 years are used and the prediction results are shown in Figure 5. The first (Figure 5a), second (Figure 5b), and third (Figure 5c) types of data are the same as the simulated data shown in Figure 2 (Up), Figure 3 (Up), and Figure 4, respectively. The first 7 years data are used as known, while the last three years data are assumed unknown for the purpose of validating the predicted results. The RMS of the differences between the simulated and predicted signals for every half year is calculated for assessing the accuracy of predicted results. As shown in Figure 6, the accuracy of the predicted results for the first and second types of data is quite stable and is around 3 mm. However, the accuracy of the predicted result from the third type of data decreases significantly after the first 1.5 years and the RMS even reaches about 10 mm in the last half 1.5 year.

4. GPS Result

4.1. Interpolation and Gross Error Detection

In practice, data gaps and gross errors inevitably exist in GPS time series, so a preprocessing procedure for missing data interpolation and gross error detection needs to be performed before a time series analysis is conducted. In our study, the SSAM (SSA for missing data) interpolation method is adopted and a new method named SSAM-IQR is used for gross error detection by combining SSAM and the interquartile range (IQR) criterion [Upton and Cook, 1996]. Due to the restriction on the length of the paper, these two methods are not discussed here as the theory can be found in several literatures [Kondrashov and Ghil, 2006; Wang et al., 2015]. The main advantage of using SSAM combined with SSM-IQR is that no prior knowledge of the time series is needed. The interpolation results for the three GPS time series used in section 4.2 for cases 1–3 are shown in Figure 7. The missing rates of these three time series are 1.1% (Figure 7a), 7.3% (Figure 7b), and 14.3% (Figure 7c).

4.2. GPS Time Series Analysis and Prediction

In this section, three GPS sites with aforementioned three types of geophysical phenomena are selected to assess SSA-PD, SSA-P, SSM, and MLE for the fitting and predicting of the GPS time series. Since the DLM

Table 4. Correlation Coefficients Between Original Data and Results Predicted From SSA-P and MLE for the Vertical Coordinates of AGMT for 2012.167–2015.178

	Original Data	SSA-P	MLE
Original data	1		
SSA-PD	0.98	1	
MLE	0.94	0.97	1

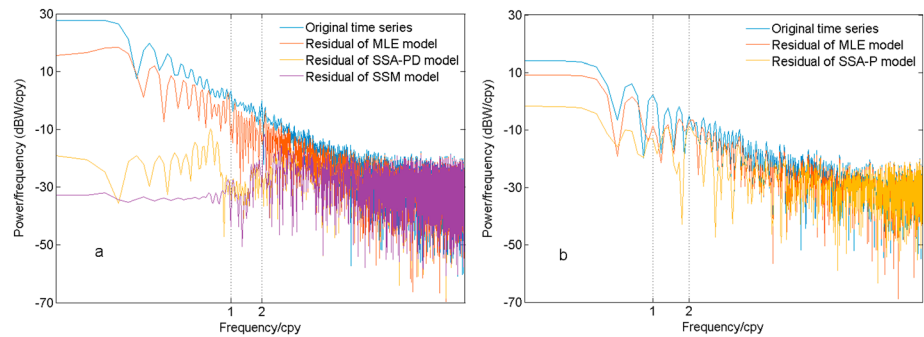


Figure 9. (a) PSD of the residuals obtained by fitting methods MLE, SSA-PD, and SSM for 1999.975–2012.167 (a) and (b) PSD of the residuals obtained by predicting methods MLE and SSA-P for 2012.167–2015.178 for the north coordinate time series of AGMT.

toolbox does not provide a prediction function, only the prediction results from SSA-P and MLE are compared. An experiment to compare the difference between the linear trends estimated by Plate Boundary Observatory (PBO) and SSA nonsecular model for 16 GPS time series is also conducted and shown in section 4.2.4.

It should be noted that the information of offset epochs provided by the SOPAC is used as a priori information in the CATS software when MLE is used to extract the trend and annual and semiannual periodic components. However, this offset information is not used in our testing of the SSA-PD and SSM methods. As stated above, for SSA-PD a 3 year lag window is applied to the original time series to extract the long-term trend first and then a 2 year lag window is applied to the detrended time series to extract the seasonal signals. When SSA-P is used for predicting the coordinates of the GPS sites, a 3 year lag window is adopted since both the trend and seasonal signals are of interest for time series analysis.

4.2.1. Case 1: GPS Time Series With a Time-Varying Trend

In the first example, the north coordinate time series of the AGMT site with a time span of 15 years is selected. AGMT is located on the North American tectonic plate with coordinates 34.59°N and 116°W. Its data missing rate is 1.1% (61 of 5554 days). It can be found from Figure 8 that the trend of its time series is noticeably not linear and two offsets (at epochs 2010.260 and 2011.293) can be observed. SSA-PD, SSM, and MLE are applied to extract the trend and periodic components from the time series between 1999.975 and 2012.167. SSM and MLE are used to extract annual and semiannual periodic components, while SSA-PD is used to extract those periodic components that have been detected from the time series. Then the daily coordinates of the AGMT site between 2012.167 and 2015.178 are predicted using both MLE and SSA-P, and the predicted results are compared with the true (observed) values for validation.

Figure 8 also shows that both SSA-PD and SSM outperform MLE, and the later totally fails in the fitting of the time series containing a time-varying trend. The RMS of the differences between the two signals (trend and periodic components) extracted from SSA-PD and SSM is 0.1 mm, which is significantly smaller than that

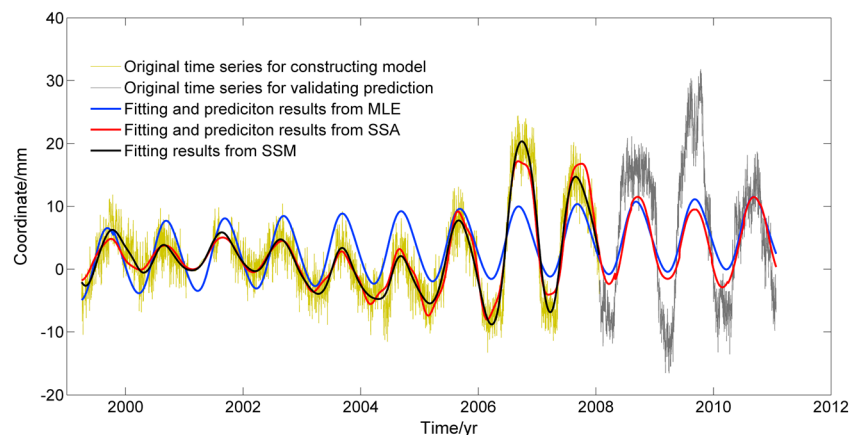


Figure 10. Signal model constructed using MLE, SSA-PD, and SSM for the vertical coordinate of DSHS for 1999.251–2011.066.

Table 5. Correlation Coefficients Between the Original Data and Fitting Results of SSA-PD, SSM, and MLE for the Vertical Coordinates of DSHS for 1999.251–2008.055

	Original Data	SSA-PD	SSM	MLE
Original data	1			
SSA-PD	0.90	1		
SSM	0.92	0.97	1	
MLE	0.64	0.72	0.70	1

between SSA-PD and MLE (4.4 mm), and also that between SSM and MLE (4.4 mm). The predicted coordinates from SSA-P also show a better agreement with the original time series than that from MLE. The RMS of the differences between the results predicted by SSA-P and MLE is 6.5 mm.

Table 3 lists the correlation coefficients between the original time series and the fitting results from SSA-PD, SSM, and MLE for the period of 1999.975–2012.167. It shows that the results from SSA-PD and SSM have the same correlation coefficient (0.99) with respect to the original time series, which is larger than that of MLE (0.96). Table 4 shows the correlation coefficients between the original time series and the results predicted by SSA-P and MLE for the period of 2012.167–2015.178. It demonstrates that the results predicted by SSA-P have a higher correlation efficient (0.98) with respect to the original data than that from MLE (0.94).

Figure 9a shows the power spectrum densities (PSD), calculated using the Lomb-Scargle algorithm [Scargle, 1982], of the original time series and the residuals of the aforementioned three methods for the period of 1999.975–2012.167. Figure 9b is the PSD of the original time series, and the residuals after SSA-P and MLE are applied for the period of 2012.167–2015.178. For the fitting results, SSA-PD and SSM significantly outperform MLE in extracting the annual and semiannual periodic components. For the prediction results (Figure 9 b), the power of the annual and semiannual components remaining in the residuals of MLE is still larger than that in the residuals of SSA-P.

4.2.2. Case Two: Time Series Containing Annual Cycle With Time-Varying Amplitude

The second case is for the vertical coordinate time series of the DSHS site with a time span of 12 years. DSHS is located on the Pacific tectonic plate with coordinates of 34.0°N, 118.3°W. The data missing rate is 7.3% (314 of 4316 days). The amplitude of its periodic components is noticeably time varying during the whole period, as shown in Figure 10. The procedure described in section 3.2.1 is performed again to obtain the fitting results of SSA-PD, SSM, and MLE for the periods of 1999.251–2008.055 and also to obtain the prediction results of SSA-P and MLE for the period of 2008.055–2011.066. Both fitting and prediction results are shown in this figure. The RMS of the differences between the fitting results of SSA-PD and SSM is 1.3 mm, which is significantly smaller than that between SSA-PD and MLE (4.0 mm) and also that between SSM and MLE (4.2 mm). The predicted coordinates from SSA-P and MLE show a similar agreement with the original time series since the RMS of the differences between these two predicted results is only 1.6 mm.

Table 5 lists the correlation coefficients between the original time series and the fitting results from SSA-PD, SSM, and MLE for the period 1999.251–2008.055. It can be seen that the fitting results from SSA-PD and SSM have the similar correlation coefficient with respect to the original time series (0.90 and 0.92, respectively), which are significantly larger than that of MLE (0.64). Table 6 shows the correlation coefficients between the original data and the results predicted by SSA-P and MLE for the period 2008.055–2011.066. It indicates that the MLE result has a slightly higher correlation coefficient than the SSA-P result.

Figure 11a shows the PSD of the original time series and the PSD of the residuals after SSA-PD, SSM, and MLE are applied to the original time series for the period of 1999.251–2008.055. Figure 11b is the PSD of the

Table 6. Correlation Coefficients Between the Original Data and Results Predicted From SSA-P and MLE for the Vertical Coordinates of DSHS for 2008.055–2011.066

	Original Data	SSA-PD	MLE
Original data	1		
SSA-PD	0.82	1	
MLE	0.84	0.97	1

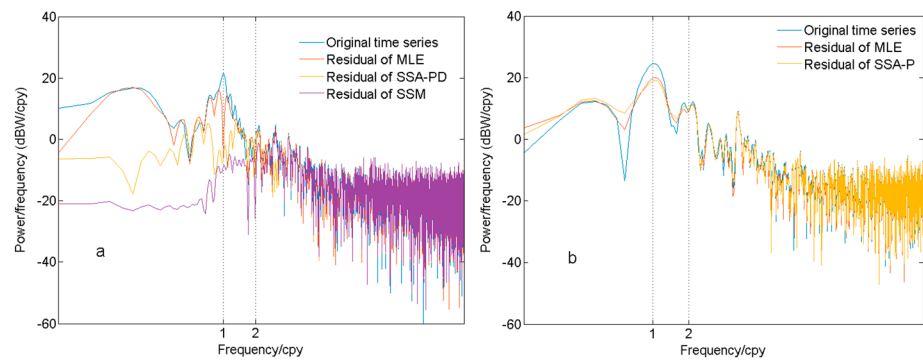


Figure 11. (a) PSD of the residuals obtained by fitting methods MLE, SSA-PD, and SSM for 1999.251–2008.055 and (b) PSD of the residuals obtained by predicting methods MLE and SSA-P for 2008.055–2011.066 for the vertical coordinate time series of DSHS.

original time series, and the PSD of the residuals after SSA-P and MLE are applied to the original time series for the period of 2008.055–2011.066.

For the fitting result, the power of the annual and semiannual components are all reduced by these three methods, compared with the original time series. However, even after the annual components are removed, there is still noticeable power around 1 cpy (cycle per year) in the residual of the MLE result. This remaining near-seasonal power could result from the failure to fit the seasonal signals with time-varying amplitudes using MLE [Bennett, 2008; Santamaría-Gómez *et al.*, 2011]. For the prediction result, the power of the annual components in the residual of MLE is also slightly larger than that of SSA-P. Since the RMS value of the differences between these two prediction results is only 1.6 mm, the accuracies of the prediction from MLE and SSA can be regarded as similar for this case study.

4.2.3. Case 3: Time Series Containing Large Offsets

The third site, KELY, is located on the North American tectonic plate (Greenland) with coordinates of 67.0°N, 50.9°W. The data missing rate is 14.3% (1021 of 7119 days). According to an analysis report given by the SOPAC, three offsets are observed in the period 1995.671–2015.159 and the largest one (−70.5 mm) is observed at the epoch 2001.707. These offsets estimated by SOPAC are first subtracted from the original time series, and CATS is then used to estimate the linear trend and the annual/semiannual components from the resultant time series. The time series analysis approach to estimating the offsets used by SOPAC has gained a great success in deformation field analysis, in particular, for coseismic fields, seasonal deformation fields, and postseismic deformation fields [Dong *et al.*, 2002]. The fitting result for 1995.671–2012.150 and prediction result for 2012.150–2015.159 are shown in Figure 12. The RMS of the differences between the fitting results of SSA-PD and SSM is 1.7 mm, which is significantly smaller than that between SSA-PD and MLE (10.6 mm) and also that between SSM and MLE (10.0 mm). The coordinates predicted from SSA-P show a better

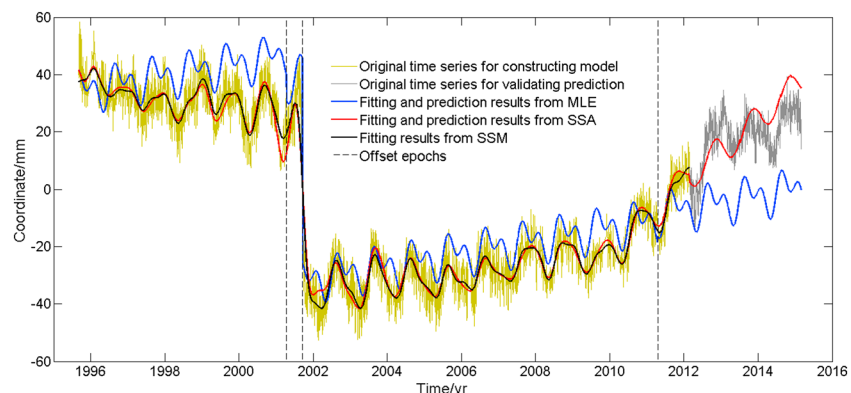


Figure 12. Signals constructed using MLE, SSA-PD, and SSM for the vertical coordinate of KELY for 1995.671–2015.159.

Table 7. Correlation Coefficients Between the Original Data and Fitting Results From SSA-PD, SSM, and MLE for the Vertical Coordinates of KELY for 1995.671–2012.150

	Original Data	SSA-PD	SSM	MLE
Original data	1			
SSA-PD	0.98	1		
SSM	0.98	0.99	1	
MLE	0.96	0.96	0.97	1

agreement with the original time series than that of MLE. The RMS of the differences between the results predicted by SSA-P and MLE is 25.3 mm.

Table 7 shows the correlation coefficients between the original time series and the fitting results from SSA-PD, SSM, and MLE for 1995.671–2012.150. One can see that SSA-PD and SSM have the same correlation coefficient (0.98) with respect to the original time series, which is larger than that of MLE (0.96). The correlation coefficients in Table 8 demonstrate that the prediction results from SSA-P (with respect to original time series) have a significantly larger correlation efficient than that of MLE.

Figure 13a shows the power of the original time series and the residual of the fitting result after these three methods are applied to the time series for the period 1995.671–2012.150, and Figure 13b shows the power of the original time series and the residuals of the prediction result after MLE and SSA-P are applied to the time series for the period 2012.150–2015.159. From the fitting result, SSA-PD and SSM significantly outperform MLE in extracting the annual and semiannual periodic components. From the predicted result, the power of the annual and semiannual periodic components remaining in the residuals of the MLE result is also larger than that of SSA-P.

It is interesting that the frequencies of the annual and semiannual components reconstructed by SSA-PD at these three sites are not exactly 1.0 cpy and 2.0 cpy, respectively. For example, the frequencies of the annual signals reconstructed from the above three time series are 1.06 cpy, 1.02 cpy, and 0.97 cpy. Given the frequency resolution (~ 0.06 cpy), these peaks could be at either the climatic (1.0 cpy) or draconitic annual frequency (1.04 cpy). Several studies [Collilieux *et al.*, 2007; Ray *et al.*, 2008; Tesmer *et al.*, 2009; Davis *et al.*, 2012; Chen *et al.*, 2013; He *et al.*, 2015] have indicated that the frequency of ~ 1.04 cpy matches the GPS constellation repeat cycle, and it could likely have resulted from some technical errors. In addition, a seasonal signal with a frequency of 0.4 cpy is also detected from the KELY time series, for which no convincing explanation can be given in this study. Several other studies also reported that low frequencies from GPS time series were also detected. Collilieux *et al.* [2007] detected significant periodic components at frequencies between 0.33 cpy and 0.36 cpy from the GPS sites located in the European area. Mtamakaya *et al.* [2012] showed at least six different groups of strong periodic components with frequencies between 0.05 cpy and 2.5 cpy detected from the time series of five sites located in Canada. Both studies did not give an explanation about where the low-frequency periodic components stemmed from.

4.2.4. Comparison Between SSA Nonsecular Model and Linear Trends From PBO

In this section, the SSA models extracted from 16 GPS time series are compared with the linear trends estimated by the PBO project. The PBO data include both the raw data collected from each instrument and processed data such as calibrated time series and velocity fields. All the time series used in this section do not contain large offsets and noticeable nonsecular trends. It can be found from Table 9 that the difference between the linear trend from PBO and the trend extracted by SSA is at a level of 1 mm at most stations in the three directions. However, the difference between the linear trend from PBO and the SSA model that contains trend and seasonal components are larger than 1 mm in the north and east directions at most stations and larger than 2 mm in the up direction at all stations. The largest RMS values of the differences in the north, east, and up direction are 7.9,

Table 8. Correlation Coefficients Between the Original Data and Predicting Results From SSA-P and MLE for the Vertical Coordinates of KELY for 2012.150–2015.159

	Original Data	SSA-PD	MLE
Original data	1		
SSA-PD	0.72	1	
MLE	0.71	0.66	1

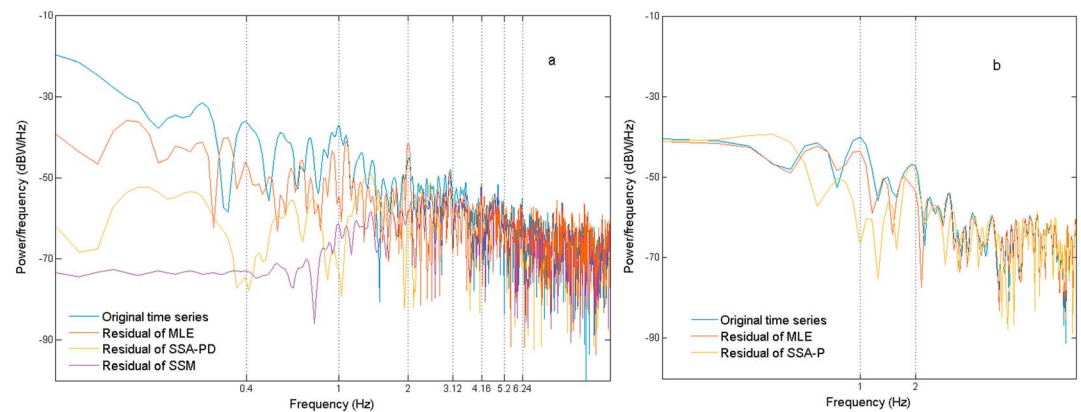


Figure 13. (a) PSD of the residuals obtained by the fitting methods MLE, SSA-PD and SSM for 1995.671–2012.150 and (b) the PSD of the residuals obtained by the predicating methods MLE and SSA-P for 2012.150–2015.159 for the vertical coordinate time series of KELY.

4.8, and 7.8 mm, respectively. This difference is not negligible when these stations are used as reference stations for those studies that require a high-accuracy TRF (e.g., 1 mm) [Gross *et al.*, 2009].

5. Conclusions

The movements of GPS sites are most likely to be nonsecular in the events of postseismic deformation or deformation associated with rapid elastic or poroelastic phenomenon. GPS time series also tend to be disrupted by offsets when equipment changes or earthquakes occur. In addition, the seasonal signals in the time series are not always perfect sinusoids with constant amplitude as assumed. In this case, instead of using a linear model, it is more meaningful to adopt a nonsecular model to depict the real movement of GPS sites, although it is challenging.

In this study, a modified method SSA-PD is proposed for solving the phase shift phenomenon (i.e., large biases in the SSA model in the beginning and end parts of a time series). Another method named SSA-P for predicting the coordinates of GPS sites is also proposed. The main advantage of SSA is that there is no need for a priori knowledge of the underlying dynamics of the time series such as the location and magnitude of offsets. The performances of these two methods in terms of time series analysis and prediction are assessed using both simulated and real GPS time series and also compared to the results obtained by MLE and SSM.

Table 9. The Difference Between SSA Model and the Linear Trends From PBO

Name	First Epoch (year)	Last Epoch (year)	Data Missing Rate (%)	RMS of the Difference Between the Linear Trend From PBO and the Trend Extracted by SSA (mm)			RMS of the Difference Between the Linear Trend From PBO and the SSA Model (Trend + Seasonal Components) (mm)		
				North	East	Up	North	East	Up
AB07	2004.8115	2014.8356	0.2	1.4	0.2	0.7	2.0	0.7	2.1
AB15	2006.6110	2014.8356	7.9	0.2	0.4	0.7	0.7	0.5	5.1
AB17	2008.9536	2014.8356	9.7	0.4	0.9	1.3	1.6	1.6	3.2
AB22	2006.4055	2014.8356	0.2	1.8	0.8	1.0	1.3	1.0	4.1
AB33	2006.7178	2014.8356	0.2	0.4	0.2	1.7	1.2	1.3	6.1
AB35	2007.6411	2014.8356	0.2	1.0	0.4	1.7	3.8	1.0	7.8
AB36	2006.6384	2014.8356	15.9	3.1	0.9	1.0	7.9	1.0	3.9
AB41	2006.7425	2014.8356	3.6	0.4	0.1	0.9	2.8	0.9	4.1
AB45	2007.4274	2014.8356	9.7	0.2	0.1	1.1	1.0	1.1	4.7
AB46	2007.2603	2014.8356	7.9	0.2	0.1	3.4	1.4	0.9	7.6
AC02	2005.7726	2014.8356	7.1	1.2	0.7	2.0	1.6	0.4	3.7
AC06	2005.6959	2014.8356	5.1	1.8	0.7	1.5	3.0	2.0	6.8
AC09	2007.6548	2014.8356	0.2	3.8	3.8	0.8	5.0	4.8	5.6
AC12	2008.3470	2014.8356	19.2	0.5	0.3	0.6	0.8	1.3	1.8
AC13	2008.7131	2014.8356	9.2	0.9	0.6	1.4	0.4	0.5	2.2
AC15	2005.7863	2014.8356	0.2	1.6	1.1	1.2	2.7	2.2	6.2

Both simulation and real GPS time series results indicate that SSA-PD can be used to analyze those GPS time series that contain nonsecular trends, offsets, and periodic components with time-varying amplitudes, and SSA-P can be used to predict their coordinates. The GPS results demonstrate that SSA-PD and SSM have a similar performance in the fitting of the GPS time series, and both outperform MLE. It is also interesting to find that the frequencies of the annual and semiannual components reconstructed by SSA-PD at three test sites are not exactly at 1.0 cpy and 2.0 cpy, respectively. These peaks could be at either the climatic or draco-nitic frequencies. In addition, a periodic component with a frequency of 0.4 cpy is reconstructed from the KELY time series by SSA-PD, meaning that taking into account only the annual and semiannual signals may not be able to fit the GPS time series to a high accuracy. The comparison between the results of the three types of time series predicted by SSA-P and MLE shows that SSA-P results in greater accuracy in the first (containing noticeable nonsecular trend) and third (containing large offsets) types of data while it has a similar performance with MLE in the second (containing seasonal components with time-varying amplitudes) type of data. However, as indicated by the simulation results, the accuracy of the results predicted by SSA-P may decrease with time when the time series contains a nonsecular trend. In this case, the coordinates of these sites are better predicted more frequently (i.e., within a short period of time, such as 1.5 years) than those sites that only contain linear trends. According to the principle of SSA, the data length used to build a SSA model is at least twice as long as the length of lag window (M) which is selected according to the property of the time series. Previous studies recommended a 3 year (or larger) lag window for reconstructing the long-term trend from a GPS time series, which means at least a 6 or 7 year data series is needed for the model construction. Therefore, more investigations on using a smaller lag window to construct a nonsecular model from a short GPS time series need to be conducted in our future studies.

In addition, the SSA models extracted from 16 GPS time series are compared to the linear trends estimated by the PBO project. The difference between the linear trend from PBO and the SSA model that contains trend and seasonal components are larger than 1 mm in the north and east directions at most stations and larger than 2 mm in the up direction at all stations. This difference cannot be neglected when constructing a high-accuracy TRF for those studies that require a highly accurate TRF with an accuracy of 1 mm.

One of the main limitations of traditional SSA is that the method used to separate signal from noise is based on the property of the RCs to distinguish corresponding trend and periodic components. This approach might be problematic if the signal-to-noise ratio is not sufficiently large, which is also mentioned in several previous studies.

Appendix A

SSA aims at making a decomposition of a time series into a number of interpretable and independent components such as a nonsecular trend and a few periodic components. Its procedure mainly consists of two stages: decomposition and reconstruction. In reconstruction stage, the number of leading RCs used to reconstruct the signal needs to be demined based on the principles proposed in [Vautard *et al.*, 1992]. We describe here the details of decomposition and reconstruction stages, Kendall test for determining the set of RCs for reconstructing trend, and two criteria for determining the set of RCs for reconstructing periodic components.

A1. Decomposition and Reconstruction

The process of decomposition consists of two steps: an embedding step and a singular value decomposition step. Let $\mathbf{X} = (x_1, \dots, x_N)$ be the original time series of length ($N > 2$). In the embedding step, $\mathbf{X}$ is mapped into a trajectory matrix $\mathbf{D}$ by sliding an M point ($1 < M < N$) lag window. $\mathbf{D}$ is formed by $L = N - M + 1$ lagged vectors $\mathbf{D}_i$:

$$\mathbf{D}_i = (x_i, \dots, x_{i+M-1})^T \quad (1 \leq i \leq L) \quad (\text{A1})$$

and

$$\mathbf{D} = [\mathbf{D}_1, \mathbf{D}_2, \dots, \mathbf{D}_{L-1}, \mathbf{D}_L] = \begin{bmatrix} x_1 & x_2 & x_3 & \cdots & \cdots & x_L \\ x_2 & x_3 & x_4 & \cdots & \cdots & \cdot \\ x_3 & x_4 & x_5 & \cdots & \cdots & \cdot \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \cdot & \cdot & \cdot & \cdot & \cdot & x_{N-1} \\ x_N & \cdot & \cdot & \cdot & x_{N-1} & x_N \end{bmatrix} \quad (\text{A2})$$

It can be seen that both rows and columns of $\mathbf{D}$ are the subseries of the original time series $\mathbf{X}$.

$\mathbf{D}$ is expanded with respect to an orthonormal basis $\mathbf{E}^k$ ($1 \leq k \leq M$):

$$\mathbf{D} = \mathbf{E} \cdot \mathbf{A} = \begin{bmatrix} E_1^1 & E_1^2 & \cdots & E_1^M \\ E_2^1 & E_2^2 & \cdots & E_2^M \\ \vdots & \vdots & \ddots & \vdots \\ E_M^1 & E_M^2 & \cdots & E_M^M \end{bmatrix} \begin{bmatrix} a_1^1 & a_2^1 & \cdots & a_L^1 \\ a_1^2 & a_2^2 & \cdots & a_L^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_1^M & a_2^M & \cdots & a_L^M \end{bmatrix} \quad (\text{A3})$$

Let $\mathbf{a}^k$ be the k th row in $\mathbf{A}$, $\mathbf{a}^k = [a_1^k, a_2^k, \dots, a_L^k]$ is the projection coefficient and defined as the principal components (PCs) of $\mathbf{D}$; let $\mathbf{E}^k$ be the k th column in $\mathbf{E}$, $\mathbf{E}^k = [E_1^k, E_2^k, \dots, E_M^k]^T$ is the basis vector and defined as the empirical orthogonal functions (EOFs) of $\mathbf{D}$.

$\mathbf{E}^k$, corresponding to the k th eigenvalue λ_k , is the k th eigenvector of a covariance matrix $\mathbf{T}_x$. $\mathbf{T}_x$ has a Toeplitz structure; i.e., its diagonals are constant corresponding to equal lags:

$$\mathbf{T}_x = \begin{bmatrix} c_0 & c_1 & c_2 & \cdots & c_{M-1} \\ c_1 & c_0 & c_1 & \cdots & \cdot \\ c_2 & c_1 & c_0 & c_1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \\ c_{M-1} & \cdot & \cdot & \cdot & c_1 & c_0 \end{bmatrix} \quad (\text{A4})$$

where c_j is the covariance coefficient of $\mathbf{X}$ at lag j ($0 \leq j \leq M-1$).

Several approaches to estimate c_j have been developed [Vautard et al., 1992] and Bos et al. discussed the Toeplitz in a geodetic context [Bos et al., 2013]. In this paper, one of the most commonly used formulae called Yule-Walker [Vautard et al., 1992; Broomhead and King, 1986] is applied:

$$c_j = \frac{1}{N-j} \sum_{i=1}^{N-j} x_i x_{i+j}, \quad 0 \leq j \leq M-1, \quad (\text{A5})$$

where N is the number of sample data points.

$\mathbf{a}^k$ is the filtered result of $\mathbf{X}$ and its i th element can be derived from equation (A3):

$$a_i^k = D_i \cdot \mathbf{E}^k = \sum_{j=1}^M x_{i+j} E_j^k, \quad 0 \leq i \leq N-M \quad (\text{A6})$$

where $\mathbf{E}^k$ needs to be sorted in descending order of λ_k .

At this stage the decomposition process has been completed. Then the reconstruction can be implemented by calculating the elements of the k th reconstruct component (RC) as proposed by Vautard et al. [1992]:

$$\text{RC}_i^k = \begin{cases} \frac{1}{M} \sum_{j=1}^M \sum_{k \in A} a_{i-j}^k E_j^k, & (M \leq i \leq N-M+1) \\ \frac{1}{j} \sum_{j=1}^j \sum_{k \in A} a_{i-j}^k E_j^k, & (1 \leq i \leq M-1) \\ \frac{1}{N-i+1} \sum_{j=i-N+M}^M \sum_{k \in A} a_{i-j}^k E_j^k, & (N-M+2 \leq i \leq N) \end{cases} \quad (\text{A7})$$

In the SSA theory, the first several leading RCs represent the trend and periodic components, and the remaining RCs are for noise in $\mathbf{X}$. Therefore, only those first several leading RCs are used to reconstruct the SSA model by superposing them together:

$$\text{SSA}_{\text{model}} = \sum_{s=1}^p \text{RC}^s \quad (\text{A8})$$

where p is the number of the leading RCs.

The procedure of determining the number of leading RCs used to reconstruct the signal is in fact the procedure of identifying the set of RCs representing the trend and periodic components. Vautard et al. [1992] applied a nonparametric test named Kendall on each PC to identify whether its corresponding RC should be used to reconstruct the trend component. They also proposed two criteria based on the spectral properties of the eigenvectors for the periodic components identification.

A2. Kendall Test for Trend Identification

Let T_i ($1 \leq i \leq n$) be a time series, calculating the number K_r of the pairs of indices (i, j) that satisfies the equation $T_i < T_j$ ($i < j$). Generally, there is a positive trend in the time series if K_r is large and a negative

trend if K_r is small. More precisely, when there is no obvious trend component in T_i , the distribution of the following coefficient

$$\tau = \frac{4K_r}{n(n-1)} - 1 \quad (\text{A9})$$

tends to be a normal distribution with a zero mean and the standard deviation of

$$s = \sqrt{\frac{2(2n+5)}{9n(n-1)}}. \quad (\text{A10})$$

Therefore, the hypothesis of no significant trend is rejected when the τ value is outside the interval $(-1.96s, +1.96s)$, with a confidence level of 95%.

The above hypothesis test is applied to the successive PCs to identify those RC candidates that represent the trend component, and then the trend component is reconstructed by superposing these identified RCs together.

A3. Two Criteria for Periodic Components Identification

Generally, when a periodic component exists in a time series, a pair of nearly equal eigenvalues always stands out of the eigenvalue series and its corresponding eigenvectors and PCs are in quadrature. However, in practice, even for a pure sinusoid, its two corresponding eigenvalues are not always equal. This means that it is indeed very difficult to identify a periodic component through searching a pair of equal eigenvalues from the eigenvalue series, which is also named eigenvalue spectrum. *Vautard et al.* [1992] proposed the following two criteria based on the spectral properties of the eigenvectors for identifying a pair of eigenvectors corresponding to a periodic component. (1) The first criterion was developed based on the fact that the periodic pairs of eigenvectors $(k, k+1)$ associated with a periodic component have the same frequency. To calculate the frequency f_k of each RC, the square of the Fourier transform $|E^k(f)|^2$ of the k th is introduced as

$$\tilde{E}^k(f) = \sum_{j=1}^M E_j^k \exp(2\pi i j f) \quad (\text{A11})$$

where $i = \sqrt{-1}$.

Then the f_k can be obtained by calculating the f corresponding to the maximum value of $|E^k(f)|^2$. If the k th and $(k+1)$ th RCs represent a periodic pair, the difference $\delta f_k = |f_k - f_{k+1}|$ has to be small enough. *Vautard et al.* [1992] suggested that the criterion with $2M \cdot \delta f_k < 0.75$ should be applied; (2) the second criterion is that if the presumed periodic pair completely represents a frequency f^* between f_k and f_{k+1} , according to the SSA theory [*Vautard et al.*, 1992], the criterion should satisfy the following equation:

$$\frac{1}{M} \left(|\tilde{E}^k(f^*)|^2 + |\tilde{E}^{k+1}(f^*)|^2 \right) = 1 \quad (\text{A12})$$

In practice, when the value of the left-hand side of the equation is larger than $\frac{2}{3}$, the criterion is regarded as being satisfied [*Vautard et al.*, 1992].

The above hypothesis test is applied to the successive PCs to identify those RC candidates that represent the periodic component, and then the trend component is reconstructed by superposing these identified RCs together.

Appendix B

B1. SSA-PD

The procedure of SSA-PD is shown in Figure B1. First, the original time series y_{t_i} ($1 \leq i \leq N$) is expanded into Y_{t_i} ($1 \leq i \leq N+2l$) by adding pseudo data (y_{pes1} and y_{pes2}) with a length of l to both beginning and end of the original time series y_{t_i} . Since the phase shift phenomenon appears only within the first and last M points of the reconstructed signal, the value of l is set to M in this study. As a result, when the expanded time series Y_{t_i} is analyzed using SSA, the phase shift will appear only within the pseudo data. The pseudo data is simulated by applying the least squares fitting to the y_{t_i} which is assumed to contain

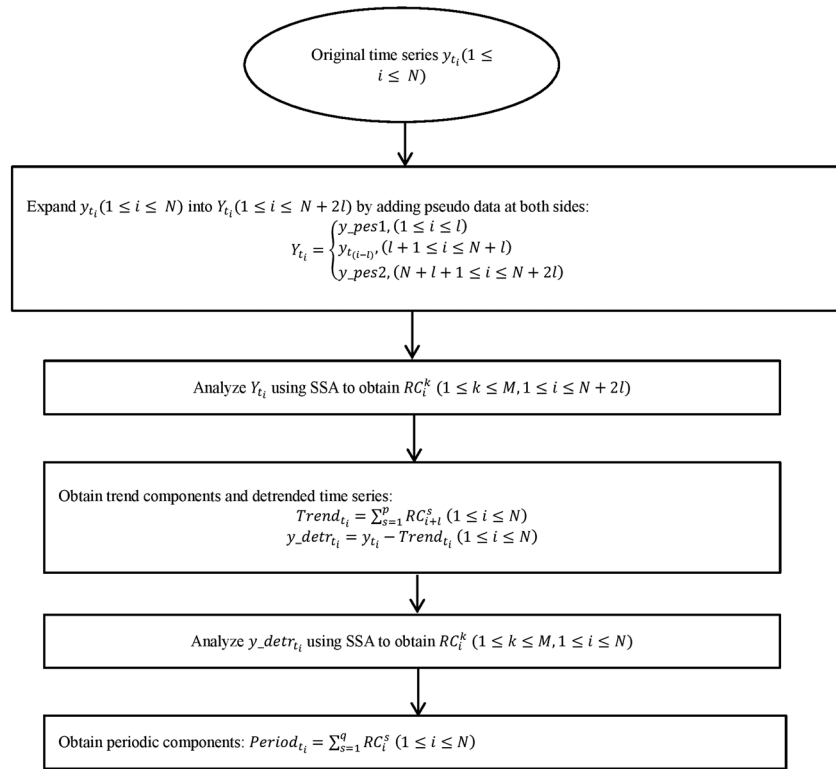


Figure B1. Procedure of SSA-PD.

a trend, the annual and semiannual periodic components. Second, after the RC_i^k ($1 \leq k \leq M$, $1 \leq i \leq N + 2l$) is obtained by analyzing Y_{t_i} using SSA, the trend component of the original time series y_{t_i} is reconstructed using

$$\text{Trend}_{t_i} = \sum_{s=1}^p RC_{i+l}^s \quad (1 \leq i \leq N) \quad (\text{B1})$$

where p is the number of RCs used in reconstructing the trend component.

The Kendall test method is used in this research to determine the value of p . The details of this test method are given in Appendix A2. Finally, the original time series y_{t_i} is detrended and the resultant series $y_{detr_{t_i}}$ ($1 \leq i \leq N$) is analyzed using SSA. The periodic components are reconstructed using

$$\text{Period}_{t_i} = \sum_{s=1}^q RC_i^s \quad (1 \leq i \leq N), \quad (\text{B2})$$

where q is the number of RCs used to reconstruct the periodic components.

Two criteria for determining the value of q are detailed in Appendix A3.

B2. SSA-P

The procedure performing SSA-P is given in Figure B2. The first step is to use SSA-PD to analyze the original time series y_{t_i} ($1 \leq i \leq N$) and reconstruct the signal s_{t_i} ($1 \leq i \leq N$). The second step is to expand s_{t_i} into yp_{t_i} ($1 \leq i \leq N + n1$) by adding $n1$ samples, $yp_initial_{t_i}$ ($1 \leq i \leq n1$), to the ending of s_{t_i} . The method to simulate $yp_initial$ is the same as the one used to simulate the pseudo data in SSA-PD illustrated in section B1. Third, the yp_{t_i} ($1 \leq i \leq N + n1$) is analyzed using SSA-PD and the reconstructed RC_i^k ($1 \leq k \leq M$, $1 \leq i \leq N + n1$) is used to update yp_{t_i} ($N + 1 \leq i \leq N + n1$) using the formula

$$yp_{t_i} = \sum_{s=1}^q RC_i^s \quad (N + 1 \leq i \leq N + n1), \quad (\text{B3})$$

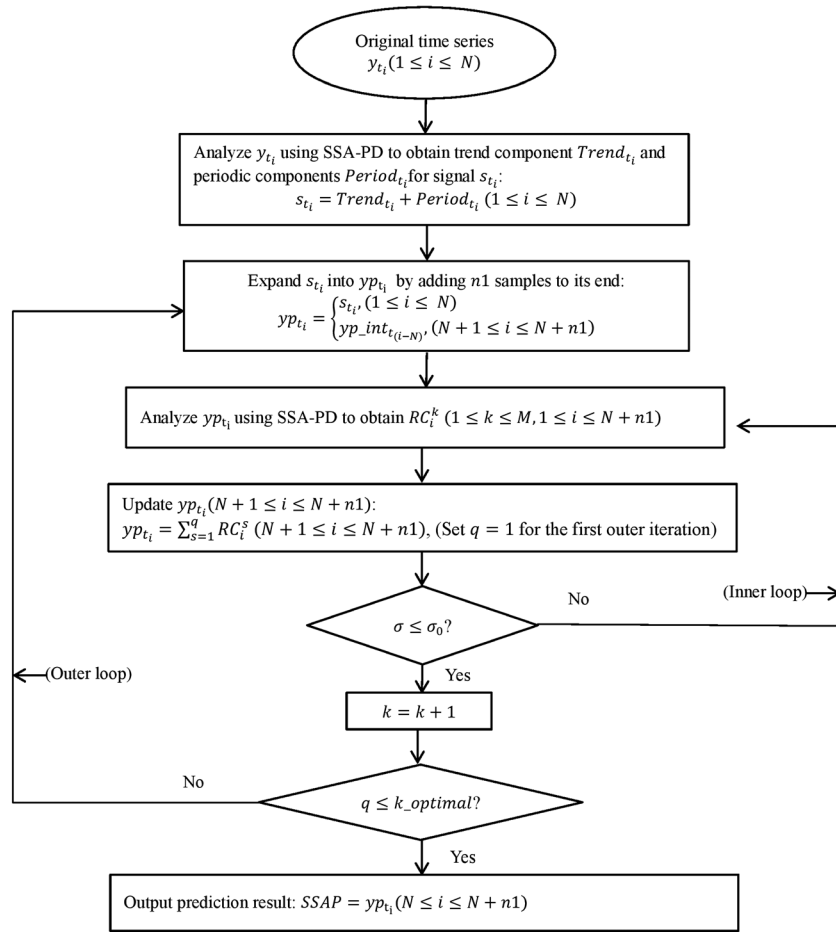


Figure B2. Procedure of SSA-P.

where q is the number of leading RCs used for updating the $yp_{t_i} (N+1 \leq i \leq N+n1)$. Its initial value is set to 1 for the first outer iteration, as indicated in Figure B2. The process (inner iteration) is repeated until the following convergence criterion is satisfied:

$$\sigma = \text{RMS}[yp_{t_i}(r+1) - yp_{t_i}(r)] \leq \sigma_0, (N+1 \leq i \leq N+n1) \quad (\text{B4})$$

where $yp_{t_i}(r+1)$ and $yp_{t_i}(r)$ are the results of current and previous inner iteration, respectively.

An appropriate value for the convergence threshold σ_0 should be selected. A large value will decrease the accuracy of the prediction result, while a small value will increase the number of iterations. In this study $\sigma_0 = 0.05$ mm is adopted, which is based on our experiment results for obtaining a high-accuracy result at low computational cost.

After the first outer iteration ($q=1$) is completed, the first two leading RCs are applied in the second outer iteration to update the prediction result $yp_{t_i} (N \leq i \leq N+n1)$. The outer iteration will be terminated when the criterion $q \leq k_{\text{optimal}}$ is satisfied and then the prediction results are obtained using $\text{SSAP} = yp_{t_i} (N \leq i \leq N+n1)$. To determine the value of k_{optimal} , a test needs to be conducted. For this test, the signal $s_{t_i} (1 \leq i \leq N)$ needs to be obtained by analyzing y using SSA-PD first. Then, the last $n1$ samples of s_{t_i} are assumed missing and the $s_{t_i} (1 \leq i \leq N-n1)$ is used to predict the missed part $s_{t_i} (N-n1+1 \leq i \leq N)$. We set $k_{\text{optimal}} = 1, 2 \dots 20$ to calculate 20 corresponding prediction results. Then 20 RMS values of the differences between the true values $s_{t_i} (N-n1+1 \leq i \leq N)$ and the 20 sets of obtained prediction results are calculated. The k_{optimal} value corresponding to the smallest RMS will be adopted in the criterion $q \leq k_{\text{optimal}}$ for the outer iteration.

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